

MANNING AGDM, AB, Canada

WMO: 712290

Lat: 56.9739N

Lon: 117.4508W

Elev: 457

StdP: 95.95

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.6	-32.8	-40.6	0.1	-36.5	-36.5	0.1	-32.7	12.1	2.2	10.4	0.9	1.7	190	0.689

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	28.1	17.2	26.3	16.5	24.6	15.7	19.1	25.3	17.9	24.1	16.9	22.8	4.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.7	12.6	21.7	15.5	11.6	20.7	14.5	10.9	19.5	56.2	25.4	52.4	24.0	49.2	22.7	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.0	7.0	-40.4	31.3	3.5	1.3	-42.9	32.3	-45.0	33.0	-46.9	33.8	-49.4	34.7		
(5)			-40.4	21.0	3.1	2.0	-42.6	22.4	-44.4	23.6	-46.2	24.7	-48.5	26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	1.0	-15.4	-13.6	-7.6	3.2	9.8	14.5	16.5	14.6	9.6	2.7	-8.8	-14.8	(6)
(7)		DBStd	13.33	9.33	8.28	7.98	4.94	4.39	2.94	2.53	3.21	4.30	5.06	7.60	8.72	(7)
(8)		HDD10.0	3885	787	662	546	208	57	3	0	3	58	231	564	767	(8)
(9)		HDD18.3	6357	1045	895	804	454	264	117	68	120	264	486	814	1026	(9)
(10)		CDD10.0	591	0	0	0	4	51	139	201	147	45	4	0	0	(10)
(11)		CDD18.3	21	0	0	0	0	1	3	11	5	1	0	0	0	(11)
(12)		CDH23.3	662	0	0	0	5	52	130	257	171	45	2	0	0	(12)
(13)		CDH26.7	124	0	0	0	1	9	21	49	37	8	0	0	0	(13)
(14)	Wind	WSAvg	3.1	2.9	2.8	3.2	3.6	3.6	3.2	2.9	2.8	3.2	3.3	2.9	2.9	(14)
(15)	Precipitation	PrecAvg	413	24	16	19	19	41	74	70	48	36	22	23	21	(15)
(16)		PrecMax	540	56	35	52	47	91	135	129	125	104	61	54	71	(16)
(17)		PrecMin	251	5	4	2	1	16	19	27	18	0	4	5	2	(17)
(18)		PrecStd	91	14	8	11	11	18	35	26	27	21	13	15	15	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.5	6.0	10.1	23.2	27.9	29.2	30.2	30.1	27.9	21.7	10.8	5.7	(19)
(20)			MCWB	3.2	2.6	5.1	10.9	14.3	16.7	19.5	18.2	16.5	12.2	6.1	3.0	(20)
(21)		2%	DB	4.5	3.8	7.2	17.6	24.4	26.5	28.0	27.3	24.1	16.1	6.9	3.1	(21)
(22)			MCWB	2.1	1.2	3.2	8.3	12.1	15.7	18.2	17.5	14.5	8.9	3.4	0.8	(22)
(23)		5%	DB	2.4	1.2	5.4	14.5	22.4	24.6	26.2	25.0	20.8	13.2	4.2	0.9	(23)
(24)			MCWB	0.7	-0.5	2.2	6.8	11.2	15.0	17.7	16.4	12.8	7.4	1.7	-0.8	(24)
(25)		10%	DB	-0.8	-1.4	3.6	12.2	20.2	22.9	24.4	23.1	18.3	10.6	1.3	-1.7	(25)
(26)			MCWB	-2.3	-2.9	1.1	5.8	10.4	14.2	16.8	15.6	11.6	6.1	-0.1	-2.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	3.0	5.4	11.3	15.2	19.0	21.7	20.0	17.2	12.9	6.4	3.2	(27)
(28)			MCDB	6.1	5.4	9.5	21.4	25.1	25.0	27.4	27.4	25.4	21.1	10.3	5.5	(28)
(29)		2%	WB	2.2	1.5	3.5	9.0	13.2	17.1	19.7	18.5	15.3	9.6	3.7	1.1	(29)
(30)			MCDB	4.3	3.6	6.6	16.4	22.2	23.7	25.9	24.8	22.1	14.8	6.4	2.9	(30)
(31)		5%	WB	0.7	-0.6	2.4	7.3	12.1	16.0	18.5	17.5	13.7	7.9	1.7	-0.6	(31)
(32)			MCDB	2.3	1.2	5.1	13.6	20.2	22.4	24.6	23.6	20.1	12.3	3.8	1.0	(32)
(33)		10%	WB	-2.3	-3.0	1.2	6.0	10.9	15.0	17.4	16.3	12.1	6.4	0.0	-2.9	(33)
(34)			MCDB	-0.8	-1.5	3.4	11.5	18.4	21.1	23.1	22.0	17.1	10.1	1.3	-1.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.0	11.9	12.7	12.3	15.2	13.8	14.0	14.6	14.3	11.1	9.7	9.7	(35)
(36)			MCDBR	12.1	14.2	13.1	17.6	20.1	18.0	17.7	18.8	18.3	15.8	12.2	11.9	(36)
(37)			MCWBR	10.1	12.0	9.8	9.9	9.9	9.0	9.5	9.9	9.6	9.6	9.3	10.1	(37)
(38)		5% WB	MCDBR	11.9	13.8	12.4	16.5	17.7	15.2	16.1	17.2	18.2	14.7	12.0	11.8	(38)
(39)			MCWBR	10.0	11.9	9.3	9.6	9.2	8.0	9.2	9.9	10.1	9.3	9.5	10.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.236	0.253	0.287	0.333	0.346	0.362	0.364	0.353	0.312	0.288	0.243	0.219	(40)	
(41)		taud	2.265	2.325	2.327	2.367	2.413	2.390	2.404	2.420	2.546	2.565	2.409	2.203	(41)	
(42)		Ebn at Noon	680	826	879	884	892	878	868	854	850	786	681	589	(42)	
(43)		Edh at Noon	47	69	91	103	106	110	106	98	74	56	41	37	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.64	1.60	3.17	4.70	5.62	5.91	5.70	4.64	3.18	1.66	0.70	0.42	(44)	
(45)		RadStd	0.06	0.14	0.25	0.28	0.44	0.50	0.35	0.39	0.24	0.22	0.06	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (18 neighbors)	N/A	N/A	+1.84	N/A	N/A	N/A	N/A	N/A	+61	N/A

Nomenclature: See separate page